



## AKGEC SKILLS FOUNDATION

### Training Program Details

#### FAB LAB CENTRE OF EXCELLENCE

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| <b>Training/Course Title:</b> | Winter Training Program<br>Embedded / Internet Of Things                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Target Audience:</b>       | Students – B. Tech / Diploma                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Fee:</b>                   | Rs. 7000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Duration:</b>              | 40 Hrs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Module</b>                 | <ul style="list-style-type: none"> <li>▪ Introduction to Embedded Systems and IoT ecosystem</li> <li>▪ Basics of electronics: sensors, actuators, power supply, and interfacing</li> <li>▪ Microcontroller fundamentals (Arduino / ESP32 / ESP8266 / Raspberry Pi – overview)</li> <li>▪ Programming fundamentals for embedded systems (C / Arduino IDE)</li> <li>▪ Digital and analog I/O interfacing</li> <li>▪ Sensor interfacing (temperature, humidity, motion, gas, light, etc.)</li> <li>▪ Actuator control (LEDs, relays, motors, buzzers)</li> <li>▪ Communication protocols: UART, I2C, SPI</li> <li>▪ Wireless communication fundamentals: Wi-Fi, Bluetooth, MQTT</li> <li>▪ IoT architecture and cloud platforms (ThingSpeak / Blynk / Firebase – basics)</li> <li>▪ Data acquisition, monitoring, and visualization</li> <li>▪ Introduction to PCB design and basic prototyping (breadboard &amp; FABLAB tools)</li> <li>▪ IoT security fundamentals and best practices</li> <li>▪ Mini project development using Embedded Systems &amp; IoT</li> <li>▪ Testing, debugging, and documentation of IoT systems</li> </ul> |
| <b>Outcome:</b>               | <ul style="list-style-type: none"> <li>▪ Understand the fundamentals of embedded systems and IoT architecture</li> <li>▪ Identify and interface sensors and actuators with microcontrollers</li> <li>▪ Write, upload, and debug embedded programs independently</li> <li>▪ Establish wired and wireless communication between devices</li> <li>▪ Connect embedded systems to cloud platforms for real-time data monitoring</li> <li>▪ Design and implement a basic IoT-based solution</li> <li>▪ Develop problem-solving and prototyping skills using FABLAB resources</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <b>Certification:</b>       | Upon successful completion, participants will be awarded a Globally recognized joint certification from FAB LAB and AKGEC. |                                   |
| <b>Program Coordinator:</b> | <b>Name:</b> Mr. Himanshu Tripathi                                                                                         | <b>Contact number:</b> 9873812070 |
| <b>Payment Mode:</b>        |                                           |                                   |